

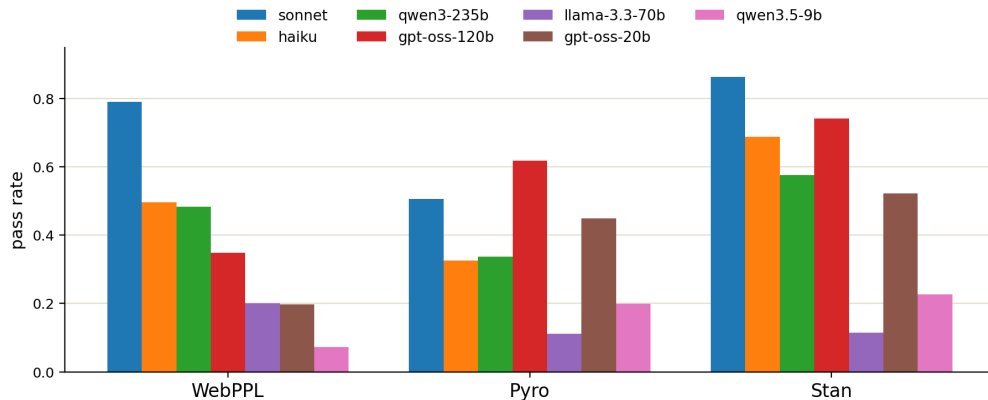
Probabilistic-programming benchmark

- **What.** Language-neutral PP problems. LLM writes the program; graded by whether its *answer* / *posterior* matches verified ground truth — not text match. Across WebPPL, Pyro, Stan.
- **Why multi-language.** PPL is low-resource — few (problem, solution) pairs. A multilingual collection lets us translate across languages, multiplying the usable corpus from the same problems.
- **The hard part.** Needs verified ground truth per language *and* one comparator that judges distributions, not strings.
- **Latest.** Ran 7 models $\times$ 3 languages (see next slide). A large share of failures trace to the *prompt*, not the model — tightening the prompt contract lifts scores substantially.



Live demo $\rightarrow$ pplgym.kingdomofends.org

Results — pass rate, 7 models × 3 languages



Fraction of problems whose answer lands inside the measured tolerance band.